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Excel

What is excel?

- * Microsoft excel is a powerful spreadsheet
- * To tools used to store, organize, analyse and data visualize data.
- * In data analytics, it acts as a beginner friendly yet powerful for handling real world datasets.

Why excel for data analytics?

- * easy to learn use.
- * handles a structure data efficiently.
- * builtin formulas and functions.
- * powerful visualization tool
- * No coding required.

Component of Excel.

1) work Book & worksheets.

- workBook → excel files (.xlsx)
- worksheet → Individual sheet inside work book.

2) Row and Columns.

- * Row → numbers (1, 2, 3, 4, ...)
- * Col → alphabets (A, B, C, D, ...)

3) Cell.

Intersection of row and columns (eg: A1)

Store the data (Text, numbers, formula)

Types of data in excel.

- * Text → names, categories.
- * numbers → sales, quantity
- * Dates → Time Based analysis.
- * Boolean → True (1) / False (0)
- * geographical data.

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Important Excel features for data analytics

1) Data cleaning / data transformation.

- * remove duplicates.

- * handling the missing values.

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- * removing the ~~irrelevant~~ irrelevant data and adding necessary data.

- * text columns

- * find and replace.

2) Formulas and function:

- * Basic $\rightarrow$ sum, average, Count
- * Logical $\rightarrow$ if, and, or, not
- * Look up $\rightarrow$ lookup, xlookup, vlookup
- * Text $\rightarrow$ concat, upper, lower

3) Data visualization:

- * charts: Bar, Col, Pie ... etc.
- * Conditional formatting
- * Sparklines

4) Pivot Tables:

- * Summarize tables
- * Group and filter data
- * Generate insights quickly

5) Sorting & Filtering:

6) Data validation:

Adv

- * Beginner friendly
- * Fast for small to medium datasets
- * widely used in companies

dis Adv

- * not suitable for very large data
- * manual errors are possible.
- * limited automation compared to coding

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Function:

- * set of instruction
 - * set of code already predefined
- is called function.

There are three ^{main components} ~~types~~ of function.

- * Function name

- * number of inputs

- * return value

1st ~~half~~ half name

Formula:

= LEFT (A15, ROUNDUP (LEN (A15) / 2, 0))

2nd half name:

= RIGHT (A15, (LEN (A15) / 2))

Upper Case:

= UPPER (A2)

* The upper case function converts all letters to upper case

Syntax:

= UPPER (Text)

Eg:

= UPPER ('Abi')

ABI

Lower Case:

The lower function is convert all letter in a text string to lower letter.

Syntax:

= LOWER (Text)

Eg:

= LOWER ('ABI')

abi

Proper Case:

The Proper function is Converts the first letter of each word, to upper Case and the remaining letters to lower Case.

Syntax:

= PROPER (Text)

Eg:

= PROPER ('ArUn')

AR Arun

Len :

The len function is a Counts the numbers of character in text string, including spaces

Syntax :

= LEN (Text)

Eg:

= LEN ('Data Analyst') $\rightarrow$ 12

Left function :

The left function extracts a specified number of character from the beginning (left side) of text string.

Syntax :

= LEFT (Text, num. chars.)

Eg:

= LEFT ('Data Analyst', 4) $\rightarrow$ Data

Right function :

The right function extracts a specified number of character from the end (right side) of text string.

Syntax :

= RIGHT (Text, <Text ->, num. chars)

Eg:

= RIGHT ("Data Analyst", 6, 7) $\rightarrow$ Analyst

MID function:

The mid function extracts a specific number of character from the middle of a text string, starting from a specified position.

Syntax: $\text{=MID}(\text{Text}, \text{start_num}, \text{num_chars})$

Eg:

$\text{=MID}(\text{"Data Analyst"}, 6, 7) \rightarrow \text{Analyst}$

Text extraction using left and len (first-half)

The left len combination is used to extract the first half of the character form a text string.

Syntax:

$\text{=LEFT}(\text{Text}, \text{LEN}(\text{Text}) / 2)$

Eg:

$\text{=LEFT}(A2, \text{LEN}(A2) / 2)$ Name: Abishek

Result: AB AB

Second-half character extraction:

The right len combination is used to extract the last half of a text string.

Syntax:

$\text{=RIGHT}(\text{text}, \text{LEN}(\text{text}) / 2)$

Eg:

If A2 contains Abishek

$\text{=RIGHT}(A2, \text{LEN}(A2) / 2)$ Result: ek

Round up function:

The Round up function rounds a number up away from zero to a specified number of digits

Syntax:

$\text{=LEFT}(\text{Text}, \text{ROUNDUP}(\text{LEN}(\text{Text})/2, 0))$

Eg: If A1 contains Abishek

$\text{=LEFT}(A2, \text{ROUNDUP}(\text{LEN}(A2)/2, 0))$

Result = Abi